# Customer segmentation methodology

Cycle 0: static K-means | Cycles 1–9: MDFCM with complete customer snapshots

## DATA PROCESSING

### 1 Data cleaning

Missing or invalid identifiers and dates;
positive paid sales; remove exact duplicates

### 2 Customer snapshots

Everyone observed before each cutoff;
exact recency and trailing 90-day purchases

### 3 RFM feature engineering

Days since purchase; invoice count and sales
in the window, expressed as monthly rates

### 4 Log transformation

log(1 + R), log(1 + F), log(1 + M);
RF, RM and FM products of logged values

### 5 MinMax scaling

Fit on Cycle 0 only; reuse the same
six-feature transformation in every cycle

### 6 Principal component analysis

Fit three components on Cycle 0 only;
reuse the same axes for fitting and plots

## CUSTOMER SEGMENTATION

### Cycle 0: K-means, K = 5

WCSS elbow selects K; silhouette supports review
Crisp assignments and initial fitted centres
Silhouette, Davies–Bouldin, CH and inertia

### Cycles 1–9: dynamic updates

Replace old snapshots with current customer records
Prior centres initialize the update
Old observations are not counted a second time
MDFCM conditions permit K−1 / K / K+1
FCM membership and centre updates
Select the eligible converged fit with lowest Xie–Beni
Track actual centres and current memberships

### Country-market analysis and integration

First five inputs log1p; all six RobustScaler
K=2–10 WCSS elbow selects 4; silhouette supports
Frozen Cycle 0 scaling and country centres
Combine market groups with behavioural profiles

## RESULTS

### Current fitted clusters

One visible entry per fitted centre
Names based on weighted RFM
Hibernating / Lost: over 90 days
Separate customer risk warnings

### Evidence and interpretation

Metrics use the fitted population
Actual PCA centre movement
K-selection and candidate results
Customer history and transactions
Market and product summaries

### Database and dashboard

Versioned SQLite and CSV outputs
Saved transforms and figures
Independent numerical audits
Local dashboard at port 4000

Reporting names and customer warnings describe results; they do not determine the learned cluster count.